

Microwave and RF Design

Wireless devices usually operate in the RF spectrum. If you are into the design of such devices, this month's CD is quite useful for you

■ NIRAJ SAHAY

Radio frequency (RF) is any frequency within the electromagnetic spectrum associated with radio wave propagation. When an RF current is supplied to an antenna, an electromagnetic field is created that is able to propagate through space.

Most wireless devices make use of RF fields. Cordless and cellular telephone, radio and television broadcast stations, satellite communication systems, and two-way radio services all operate in the RF spectrum.

Below are some of the software that can be used for microwave and RF designing.

Sonnet Lite

Sonnet Lite, a free software for high-frequency electromagnetic analysis, is a feature-limited version of Sonnet's professional Sonnet Suite. Its latest version is 12.53

What's Special?

Using Sonnet Lite, RF and microwave designers can incorporate up to three surface-mount devices (SMDs) in a circuit. The software is useful for simulation of surface-mount inductors, capacitors and resistors. The effects of the SMD terminal contact width on the circuit, as well as interaction with the surface-mount pad may be included, and this sometimes has a significant electrical effect at higher frequencies. Sonnet Lite also includes seamless interfaces to popular RF electronic design automation frameworks from Agilent (Agilent ADS) and Applied Wave Research (AWR Microwave Office). You can use it within these frameworks for modeling microwave circuits or use Sonnet's internal project editor for drawing circuits, vias, filters

or other 3D planar circuit layouts.

Distributor: Being free, it can be downloaded freely from the website of the software.

Home page: www.sonnetsoftware.com

Download URL: www.sonnetsoftware.com/products/lite/download.html

Platform Supported: Microsoft Windows

FEKO

FEKO is a comprehensive electromagnetic simulation software tool that helps you solve a variety of electromagnetic issues.

What's Special?

FEKO combines a wide range of mathematical methods for simulation of microwave and EM problems. Multiple solution techniques make it applicable to a wide range of problems for a large array of industries. The generally applicable and powerful multilevel fast multipole method and hybridisation of the methods of moments (MoM) with the finite element method and asymptotic high-frequency physical optics, geometrical optics and uniform theory of diffraction technique make FEKO particularly suitable for solving electrically large problems and thus a better proposition than traditional codes.

FEKO simulations run in parallel on all the cores of a multi-core CPU, without incurring extra licence costs.

Distributor: The software can be downloaded from the website. Request for a full 45-day free evaluation of FEKO can also be ordered on its website. Download area also provides FEKO Lite, which is free to use but has restriction on the size of problems to solve.

Home page: www.feko.info/

Download URL: www.feko.info/download

Platforms Supported: Microsoft Windows and GNU/Linux

AppCAD

AppCAD is a powerful, easy-to-use RF/microwave circuit design software. Its latest version is 3.0.2

What's Special?

AppCAD provides you with a suite of RF design tools and computerised application notes to make your design job faster and easier. It is based on the very popular MS-DOS version of Agilent Technologies' original RF and Microwave AppCAD 'Classic.'

Distributor: The software is free and can be downloaded from website.

Home page: www.hp.woodshot.com

Download URL: www.hp.woodshot.com/appcad/version302/setup.exe

Platform Supported: Microsoft Windows

Multimatch IMW

Multimatch IMW is an impedance-matching software that can be used to synthesise lumped-element, distributed or mixed lumped/distributed matching networks.

What's Special?

Multimatch IMW synthesises high-quality RF and microwave impedance-matching networks up to microstrip level. While working with the software, microstrip lines are assumed to be lossless during synthesis. Pads and connecting lines, as well as parasitic inductance for capacitors and parasitic capacitance for inductors, can be specified for the networks to be synthesised.

Distributor: The software can be downloaded from the website of the software.

Home page: www.ampsa.com

Download URL: www.ampsa.com/ftp/IMW.htm

Platform Supported: Microsoft Windows ●

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